

Undergraduate Curriculum in Biological Systems Engineering

Ecological Engineering Option

2025-2026 Catalog

Total Credits 129

First Year (33 cr.)

Semester 1 (16 cr.)

R	Engr 1010	Orientation (FS)
3	ABE 1700	Graphics and Design (FS)
4	Chem 1670	General Chemistry (FS)
1	Chem 1670L	General Chemistry Lab (FS)
4	Math 1650	Calculus I (FSSS)
3	Engl 1500	Crit. Think. and Comm. (FSSS)
1	Lib 1600	Information Literacy (FSSS)

Semester 2 (17 cr.)

1	ABE 1100	Experiencing BSE (S)
3	ABE 1600	Engineering Problems (S)
4	Math 1660	Calculus II (FSSS)
4	Phys 2310	Classical Physics I (FSSS)
1	Phys 2310L	Classical Physics I Lab (FS)
3	Biol 2120	Principles of Biology II (FSSS)
1	CE 2710	Engineering Foundations of Statics (FSSS)

Second Year (33 cr.)

Semester 3 (16 cr.)

3	ABE 2160	Fund. Ag. and Biol. Engineering (F)
2	CE 2720	Applied Engineering Statics (FS)
3	ENGL 2500	WOVE Composition (FSSS)
2	Chem 2110	Quant. & Environ. Analysis (FS)
2	Chem 2110L	Quant. & Environ. Analysis Lab (FS)
4	BIOL 3120	Ecology (FSS)

Semester 4 (17 cr.)

2	ABE 2180	Project Mgmt/Design Ag & Bio Engr (S)
1	ABE 2010	Preparing for the Workplace (FS)
3	ABE 3780	Mechanics of Fluids (FS)
1	→	Lab Elective (<i>ABE 3780L preferred</i>)
3	STAT 3050	Engineering Statistics (FSSS)
4	Math 2670	Differential Equations (FSSS)
3	→	US Cultures & Communities Elective

Third Year (32 cr.)

Semester 5 (16 cr.)

1	ABE 2730	2D Auto CAD (FS)
3	ABE 3160	Applied Numerical Methods for ABE
3	ABE 3340X	Principles of Ecological Engineering
3	CE 3720	Engr. Hydrology & Hydraulics (FS)
3	ME 2310	Thermodynamics (FSSS)
3	→	Communication Elective.

Semester 6 (16 cr.)

4	ABE 3630	Electric Power and Electronics (FS)
3	ABE 3800	Principles of BSE (S)
3	ABE 4310	Des. & Eval. of Soil & Water Cons. (F)
3	EM 3240	Mechanics of Materials (FSSS)
3	IE 3050	Engineering Economic Analysis (FSSS)

Fourth Year (31 cr.)

Semester 7 (17 cr.)

2	ABE 4150	Ag & Biosystems Engr. Design I (FS)
3	ABE 4040	Instrumentation for Ag and Bio Engr (F)
3	ABE 4340	Ecosystem Restoration (S)
3	ABE 4800	Engineering Analysis of Bio Systems (F)
3	Micro 3020	Biology of Microorganisms (FSSS)
3	→	SS/H Elective

Semester 8 (14cr.)

2	ABE 4160	Ag & Biosystems Engr. Design II (FS)
3	→	Ecological Elective I
3	→	Ecological Elective II
3	→	SS/H Elective
3	→	International Perspectives Elective

**Please check the current catalog and Schedule of Classes for most recent offerings*